

GAME THEORY, Spring 2026

Problem Set 1

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1 n -Bidder Auction

Consider the first/second price auction environment we covered in class. Instead of 2 bidders, assume there are n bidders with value $v_1, \dots, v_n$.

- (a) Extend the first price auction to this case. As usual, if more than one bidder bid the same highest price, a winning bidder is randomly drawn from them with equal probabilities. (Use this tie breaking rule also for the next question).
- (b) Extend the second price auction to this case.

Answer: We write the induced normal-form game in the standard way.

- (a) **First-price auction.** Let the set of players be $N = \{1, \dots, n\}$. For each bidder $i \in N$, the strategy space is $S_i = \mathbb{R}_+$. For a bid profile $s = (s_1, \dots, s_n) \in \prod_{i=1}^n S_i$, define the highest bid by

$$M(s) = \max_{j \in N} s_j,$$

and the set of highest bidders by

$$H(s) = \{j \in N : s_j = M(s)\}.$$

The payoff of bidder i is

$$u_i(s) = \begin{cases} \frac{v_i - s_i}{|H(s)|}, & i \in H(s), \\ 0, & i \notin H(s). \end{cases}$$

- (b) **Second-price auction.** Again the player set is $N = \{1, \dots, n\}$ and each bidder i chooses a bid $s_i \in S_i = \mathbb{R}_+$. Let

$$S(s) = \max\{s_j : j \in N \setminus H(s)\}$$

denote the highest bid among those who are not tied for the win. Then bidder i 's payoff can be written as

$$u_i(s) = \begin{cases} v_i - S(s), & \text{if } |H(s)| = 1 \text{ and } i \in H(s), \\ \frac{v_i - M(s)}{|H(s)|}, & \text{if } |H(s)| \geq 2 \text{ and } i \in H(s), \\ 0, & i \notin H(s). \end{cases}$$

2 Iterated Elimination

In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

		Player 2		
		L	C	R
Player 1	U	6, 8	2, 6	8, 2
	M	8, 2	4, 4	9, 5
	D	8, 10	4, 6	6, 7

Answer: First, for Player 1, strategy U is strictly dominated by M , since

$$8 > 6, \quad 4 > 2, \quad 9 > 8.$$

Hence U is deleted.

After removing U , consider Player 2's payoffs in the reduced game with rows $\{M, D\}$. Now column C is strictly dominated by column R , because

$$5 > 4 \quad \text{and} \quad 7 > 6.$$

Hence C is deleted.

In the remaining game with rows $\{M, D\}$ and columns $\{L, R\}$, no further strict dominance exists. Therefore, the strategy profiles that survive IESDS are

$$(M, L), \quad (M, R), \quad (D, L), \quad (D, R).$$

3 Roommates

Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j , then player i 's payoff is given

by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

- (a) What is the best response correspondence of each player i ?
- (b) Which choices survive one round of IESDS?
- (c) Which choices survive IESDS?

Answer:

- (a) Fix player j 's choice $t_j \geq 0$. Player i solves

$$\max_{t_i \geq 0} u_i(t_i, t_j) = \max_{t_i \geq 0} ((10 - t_j)t_i - t_i^2).$$

Since this objective is strictly concave in t_i , the best response is unique. The first-order condition is

$$\frac{\partial u_i}{\partial t_i} = 10 - t_j - 2t_i = 0,$$

so the unconstrained maximizer is $t_i = \frac{10 - t_j}{2}$. Imposing $t_i \geq 0$, we obtain

$$BR_i(t_j) = \max \left\{ \frac{10 - t_j}{2}, 0 \right\}.$$

Equivalently,

$$BR_i(t_j) = \begin{cases} \frac{10 - t_j}{2}, & 0 \leq t_j \leq 10, \\ 0, & t_j \geq 10. \end{cases}$$

By symmetry, player j has the same best-response correspondence.

- (b) Let $a, b \geq 0$ be two actions for player i . Then

$$u_i(a, t_j) - u_i(b, t_j) = (10 - t_j)a - a^2 - ((10 - t_j)b - b^2) = (a - b)(10 - t_j - a - b).$$

If $b > 5$, then action 5 strictly dominates b , because for every $t_j \geq 0$,

$$u_i(5, t_j) - u_i(b, t_j) = (5 - b)(5 - t_j - b) > 0,$$

since $5 - b < 0$ and $5 - t_j - b < 0$ for all $t_j \geq 0$ when $b > 5$. Hence every action $b > 5$ is eliminated. On the other hand, no action in $[0, 5]$ is strictly dominated in the original game. Indeed, if $b \leq 5$ and some $a > b$ were to strictly dominate b , we would need

$$(a - b)(10 - t_j - a - b) > 0 \quad \text{for all } t_j \geq 0,$$

which is impossible because for sufficiently large t_j the second factor becomes negative. If instead $a < b$, then strict dominance for all $t_j \geq 0$ would require $a + b > 10$, which

cannot hold when $b \leq 5$ and $a < b$. Therefore, after one round of IESDS, the surviving choices are exactly

$$[0, 5].$$

(c) Let the current surviving interval be $[L, U]$. For any $a, b \in [L, U]$,

$$u_i(a, t_j) - u_i(b, t_j) = (a - b)(10 - t_j - a - b).$$

If opponents are restricted to $t_j \in [L, U]$, then every action below $\frac{10-U}{2}$ is strictly dominated by $\frac{10-U}{2}$, and every action above $\frac{10-L}{2}$ is strictly dominated by $\frac{10-L}{2}$. Hence one more round of elimination maps the interval $[L, U]$ to

$$\left[\frac{10-U}{2}, \frac{10-L}{2} \right].$$

Starting from the one-round survivor set $[0, 5]$, we obtain the sequence

$$[0, 5] \rightarrow \left[\frac{5}{2}, 5 \right] \rightarrow \left[\frac{5}{2}, \frac{15}{4} \right] \rightarrow \left[\frac{25}{8}, \frac{15}{4} \right] \rightarrow \dots$$

These intervals converge to the unique fixed point t^* satisfying

$$t^* = \frac{10 - t^*}{2} \implies t^* = \frac{10}{3}.$$

Therefore, the only choice that survives IESDS is

$$t_i = \frac{10}{3}$$

for each player.

4 Campaigning

Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign $[N]$). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π . The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.

If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.

If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.

If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

- Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)
- Write down the game in matrix form.
- What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

Answer:

- Let the set of players be $N = \{1, 2\}$, and each player's strategy set be

$$S_i = \{P, B, N\}, \quad i \in \{1, 2\}.$$

For player 1, the payoff function $u_1 : S_1 \times S_2 \rightarrow [0, 1]$ is given by player 1's probability of winning:

$$u_1(P, P) = u_1(B, B) = u_1(N, N) = 0.5,$$

$$u_1(P, B) = 0, \quad u_1(B, P) = 1,$$

$$u_1(P, N) = 0.3, \quad u_1(N, P) = 0.7,$$

$$u_1(N, B) = 0.6, \quad u_1(B, N) = 0.4.$$

Since the game is symmetric and one candidate's probability of winning is the other candidate's probability of losing, player 2's payoff function is given by

$$u_2(s_1, s_2) = 1 - u_1(s_1, s_2).$$

- The game in matrix form is therefore

		Candidate 2		
		P	B	N
Candidate 1	P	0.5, 0.5	0, 1	0.3, 0.7
	B	1, 0	0.5, 0.5	0.4, 0.6
	N	0.7, 0.3	0.6, 0.4	0.5, 0.5

- First, for each player, strategy P is strictly dominated by N , so P is eliminated for both players. After deleting P , the reduced game has strategies $\{B, N\}$ for each player. In this

reduced game, for each player, B is strictly dominated by N . Therefore B is eliminated as well, and the only surviving strategy for each player is N . Hence the unique strategy profile that survives IESDS is

$$(N, N).$$

So in this game, IESDS does lead to a clear prediction: both candidates run negative campaigns.

5 Beauty Contest Best Responses

Consider a game with n players, so that $N = \{1, \dots, n\}$. Each player has to choose an integer between 0 and 20, so that $S_i = \{0, 1, 2, \dots, 19, 20\}$. The winners are the players who choose an integer number that is closest to $\frac{3}{4}$ of the average.

Because more than one person can win, define the set of winners $W \subset N$ as those players who are closest to $\frac{3}{4}$ of the average, and the rest are all losers. The set of winners is defined as

$$W = \left\{ i \in N : \arg \min_{i \in N} \left| s_i - \frac{3}{4} \frac{1}{n} \sum_{j=1}^n s_j \right| \right\}$$

To introduce payoffs, each player pays 1 to play the game, and winners split the pot equally among themselves. This implies that if there are $m \geq 1$ winners, each gets a payoff of $\frac{N-m}{m}$ (his share of the pot net of his own contribution) while losers get -1 (they lose their contribution).

- Show that if player i believes that everyone else is choosing 20 then 19 is not the only best response for any number of players n .
- Show that the set of best-response strategies to everyone else choosing the number 20 depends on the number of players n .

Answer:

- Suppose every player other than i chooses 20, and let player i choose $x \in \{0, 1, \dots, 20\}$. Then the average is

$$\bar{s}(x) = \frac{20(n-1) + x}{n},$$

so the target number is

$$T(x) = \frac{3}{4} \bar{s}(x) = \frac{3}{4} \cdot \frac{20(n-1) + x}{n} = 15 - \frac{15}{n} + \frac{3x}{4n}.$$

Consider first $x = 19$. Then

$$T(19) = 15 - \frac{3}{4n},$$

so player i 's distance from the target is

$$|19 - T(19)| = 4 + \frac{3}{4n},$$

while each player choosing 20 is at distance

$$|20 - T(19)| = 5 + \frac{3}{4n}.$$

Hence 19 makes player i the unique winner.

Now consider $x = 18$. Then

$$T(18) = 15 - \frac{3}{2n},$$

and therefore

$$|18 - T(18)| = 3 + \frac{3}{2n}, \quad |20 - T(18)| = 5 + \frac{3}{2n}.$$

So 18 also makes player i the unique winner.

If player i is the unique winner, his payoff is $n - 1$. Thus both 18 and 19 yield payoff $n - 1$, so 19 is *not* the only best response for any n .

- (b) Again let player i choose $x \in \{0, 1, \dots, 20\}$ while every other player chooses 20. If $x = 20$, then everyone ties as a winner, so player i 's payoff is

$$\frac{n - n}{n} = 0.$$

If $x < 20$, then the other $n - 1$ players all choose the same number, so either player i is the unique winner or else he loses. Hence for $x < 20$,

$$u_i(x, 20, \dots, 20) = \begin{cases} n - 1, & \text{if } |x - T(x)| < |20 - T(x)|, \\ -1, & \text{if } |x - T(x)| > |20 - T(x)|. \end{cases}$$

Therefore, the best responses are exactly those $x < 20$ for which player i is strictly closer to the target than 20 is.

We solve

$$|x - T(x)| < |20 - T(x)|.$$

Since $x < 20$, we have $x - 20 < 0$, so after squaring both sides and factoring,

$$(x - T(x))^2 < (20 - T(x))^2 \iff (x - 20)(x + 20 - 2T(x)) < 0 \iff x + 20 - 2T(x) > 0.$$

Substituting for $T(x)$ gives

$$x + 20 - 2 \left(15 - \frac{15}{n} + \frac{3x}{4n} \right) > 0,$$

which simplifies to

$$x \left(1 - \frac{3}{2n} \right) > 10 - \frac{30}{n} \iff x(2n - 3) > 20(n - 3).$$

Thus the set of best responses is

$$BR_i(20, \dots, 20) = \left\{ x \in \{0, 1, \dots, 19\} : x > \frac{20(n - 3)}{2n - 3} \right\}.$$

This set depends on n . For example,

$$BR_i(20) = \{0, 1, \dots, 19\} \quad \text{when } n = 2,$$

while

$$BR_i(20, 20, 20) = \{5, 6, \dots, 19\} \quad \text{when } n = 4.$$

Hence the set of best-response strategies to everyone else choosing 20 does indeed depend on the number of players n .

6 IESDS and BR in Infinite Games

Consider the following two-player game. Each player announces a nonnegative real number. The payoffs are

$$v_i(x_i, x_j) = \begin{cases} 2, & \text{if } x_i = 0 \text{ and } x_j = 1, \\ \arctan x_i, & \text{if otherwise.} \end{cases}$$

- Argue that every positive announcement is strictly dominated.
- Argue that announcement 0 is not strictly dominated.
- From the above two questions, we know only 0 survives IESDS for both players. Are they mutual best responses?

Answer:

- Fix any announcement $x_i > 0$. We claim that x_i is strictly dominated by any larger announcement $y_i > x_i$. Indeed, for every announcement $x_j \geq 0$ of player j , if player i chooses $x_i > 0$, then the special case in the payoff definition never applies, so

$$v_i(x_i, x_j) = \arctan x_i.$$

Likewise,

$$v_i(y_i, x_j) = \arctan y_i.$$

Since $\arctan(\cdot)$ is strictly increasing on $\mathbb{R}_+$ and $y_i > x_i$, we have

$$\arctan y_i > \arctan x_i.$$

Therefore,

$$v_i(y_i, x_j) > v_i(x_i, x_j) \quad \text{for all } x_j \geq 0,$$

so every positive announcement is strictly dominated.

- (b) Let $y_i \geq 0$ be any alternative announcement. If $y_i = 0$, then y_i clearly does not strictly dominate 0, since the two strategies are identical.

If $y_i > 0$, consider the opponent announcement $x_j = 1$. Then

$$v_i(0, 1) = 2,$$

while

$$v_i(y_i, 1) = \arctan y_i < \frac{\pi}{2} < 2.$$

Hence y_i cannot strictly dominate 0. Since no strategy $y_i \geq 0$ strictly dominates 0, announcement 0 is not strictly dominated.

- (c) From parts (a) and (b), the only strategy that survives IESDS is 0 for each player. However, these are *not* mutual best responses.

To see this, fix $x_j = 0$. Then for player i ,

$$v_i(x_i, 0) = \arctan x_i \quad \text{for all } x_i \geq 0,$$

because the exceptional payoff 2 occurs only when $(x_i, x_j) = (0, 1)$. Since $\arctan x_i$ is strictly increasing in x_i , we have

$$\arctan x_i > \arctan 0 = 0 \quad \text{for every } x_i > 0.$$

So $x_i = 0$ is not a best response to $x_j = 0$.

In fact, there is no best response to $x_j = 0$ at all: the payoff $\arctan x_i$ can be made arbitrarily close to $\pi/2$ by choosing larger and larger x_i , but it never reaches $\pi/2$ for any finite x_i . Thus the best-response set to 0 is empty.

Therefore, although IESDS leaves only the strategy profile $(0, 0)$, that profile does not consist of mutual best responses.

7 n -Firm Cournot Competition

Consider the n -firm Cournot competition. The demand curve is still

$$D(Q) = \max\{100 - Q, 0\}.$$

If each firm i supplies q_i , the total supply is $\sum_{i=1}^n q_i$. Suppose each firm's marginal cost is 10.

- (a) Write down its normal form game.
- (b) For each firm, what are the strategies that survive IESDS?

Answer:

- (a) Let the set of firms be $N = \{1, \dots, n\}$. Firm i chooses an output level $q_i \in S_i = \mathbb{R}_+$. Since each firm has constant marginal cost 10, firm i 's profit is

$$v_i(q_1, \dots, q_n) = q_i(D(Q) - 10) = q_i \left(\max \left\{ 100 - \sum_{k=1}^n q_k, 0 \right\} - 10 \right).$$

Equivalently,

$$v_i(q) = \begin{cases} q_i (90 - \sum_{k=1}^n q_k), & \text{if } \sum_{k=1}^n q_k \leq 100, \\ -10q_i, & \text{if } \sum_{k=1}^n q_k > 100. \end{cases}$$

Therefore the normal-form game is

$$G = (N, (S_i)_{i \in N}, (v_i)_{i \in N})$$

with $N = \{1, \dots, n\}$, $S_i = \mathbb{R}_+$ for every i , and payoff functions v_i as above.

- (b) We first derive firm i 's best response to the total output of the other firms. Let

$$x = \sum_{j \neq i} q_j.$$

Then firm i chooses $q_i \geq 0$ to maximize

$$q_i(\max\{100 - x - q_i, 0\} - 10).$$

If $x \geq 90$, every positive q_i yields negative profit, so the best response is $q_i = 0$. If $x < 90$, then the relevant objective is

$$q_i(90 - x - q_i),$$

whose unique maximizer is obtained from the first-order condition

$$90 - x - 2q_i = 0.$$

Thus

$$BR_i(x) = \max \left\{ \frac{90 - x}{2}, 0 \right\}.$$

We now perform IESDS.

First, every quantity $q_i > 90$ is strictly dominated by 0, because for every profile of the other firms,

$$v_i(q_i, q_{-i}) < 0 = v_i(0, q_{-i}).$$

Hence after the first round each firm's surviving strategies lie in

$$[0, 90].$$

Next, in this reduced game every quantity $q_i > 45$ is strictly dominated by 45. Indeed, let $x = \sum_{j \neq i} q_j \geq 0$ and take any $q_i > 45$. If $x + q_i \leq 100$, then both profits are computed in the positive-price region, and

$$v_i(45, x) - v_i(q_i, x) = (45 - q_i)(45 - x - q_i) > 0,$$

because $45 - q_i < 0$ and $45 - x - q_i < 0$. If $x + q_i > 100$, then $v_i(q_i, x) = -10q_i < -450$. On the other hand, if $x \leq 55$, then

$$v_i(45, x) = 45(45 - x) \geq -450,$$

while if $x > 55$, then also $x + 45 > 100$, so

$$v_i(45, x) = -450.$$

In either case $v_i(45, x) > v_i(q_i, x)$. Hence every quantity above 45 is eliminated. Therefore, after two rounds, each firm's surviving strategies are contained in

$$[0, 45].$$

At this point the answer depends on n .

If $n = 2$, let the current surviving interval be $[L, U] \subseteq [0, 45]$. Since the opponent's output lies in $[L, U]$, firm i 's best responses lie in

$$\left[\frac{90 - U}{2}, \frac{90 - L}{2} \right].$$

Hence one more round of elimination maps $[L, U]$ to

$$\left[\frac{90 - U}{2}, \frac{90 - L}{2} \right].$$

Starting from $[0, 45]$, we obtain

$$[0, 45] \rightarrow \left[\frac{45}{2}, 45 \right] \rightarrow \left[\frac{45}{2}, \frac{135}{4} \right] \rightarrow \dots$$

and these intervals converge to the unique fixed point q^* satisfying

$$q^* = \frac{90 - q^*}{2}.$$

Solving gives

$$q^* = 30.$$

So when $n = 2$, the only strategy that survives IESDS is

$$q_i = 30.$$

If $n \geq 3$, no further elimination is possible after reaching $[0, 45]$. To see this, fix any $q \in [0, 45]$. Then

$$x = 90 - 2q \in [0, 90].$$

Because there are at least two rival firms, we can choose their outputs in $[0, 45]$ so that their total output is exactly x (for instance, let one rival produce $\min\{45, x\}$, another produce the remainder, and any further rivals produce 0). Against such a profile, firm i 's best response is

$$BR_i(x) = \frac{90 - x}{2} = q.$$

Hence every $q \in [0, 45]$ is a best response to some surviving profile of the other firms, so no such strategy can be strictly dominated. Therefore, when $n \geq 3$, the strategies that survive IESDS are exactly

$$[0, 45].$$